Quantum screening jets and deformation intertwiners for the symmetric Carrollian W_3-algebra

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Abstract

We construct the simple-root quantum screening jets of the proper symmetric Carrollian W_3-algebra obtained from the contracted Miura transformation. For each simple root, the two relativistic Fateev–Lukyanov screening families contract to one such jet. Its leading coefficients $Q_i^{[0]}$ are finite, non-zero intertwiners on completed horizontal Fock families and screen the spin-two and primary spin-three generators precisely on $\alpha^+\mu = 1$. The parent reciprocity $\beta_+\beta_- = -4$ becomes $\alpha^+\beta_1 = -2$: one branch remains a local screening, whereas the reciprocal branch is absorbed into the background charge. The first relative coefficients are also finite and non-zero, but their target representations have different first tangents. In a fixed PBW frame the canonical coefficients $Q_i^{[1]}$ therefore satisfy an inhomogeneous deformation-intertwiner equation, rather than an ordinary commutant relation. We verify its nonlinear spin-three form directly from the two finite-parent orderings; their average supplies a fixed contact term, and no additional counterterm is introduced. Finally, on the gravity branch $\alpha^- = 0$, $\alpha^+ = \mathrm{i}a_s$, we establish the semiclassical dictionary to flat higher-spin gravity. The bulk leading and barred-linear Noether charges realize $Q_i^{[0],\mathrm{cl}}$ and the raw coefficient $\widetilde{Q}_i^{[1],\mathrm{cl}}$; the canonical $Q_i^{[1]}$ is the corresponding PBW covariantization and is not an ordinary second screening.

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1 Introduction

Free-field realizations organize the nonlinear structure of two-dimensional W-algebras [22] in a form adapted to both representation theory and Coulomb-gas methods. In the relativistic theory, the quantum Miura transformation packages the currents into an ordered differential operator, while screening charges intertwine Fock modules and characterize the W-currents through commutant conditions [4–8, 10–12]. It is therefore natural to ask what survives under a Carrollian contraction.

For three-dimensional flat holography, this question is particularly concrete. The asymptotic symmetry algebra contains $\mathfrak{bms}_3$, interpreted as a conformal Carroll algebra [17–19]. Fredenhagen and Hörl constructed proper symmetric quantum Carrollian W_N-algebras by contracting two relativistic Miura operators [1]. The reversal of the barred polarization forces a symmetric average of the two parent orderings; with that prescription, the basic structure constants retain their classical form; earlier free-field realizations of $\mathfrak{bms}_3$ and its extensions were studied in [27]. Building on the Brown–Henneaux analysis, W-algebras arise as asymptotic symmetries in AdS_3 higher-spin gravity [16, 20, 21], while their flat spin-three counterparts and flat higher-spin phases were developed in [23–25]; see [26] for a systematic account. Independently, D'Arcy, Delfante and Fredenhagen derived a classical Carrollian Coulomb gas from asymptotically flat higher-spin Chern–Simons gravity in a flat diagonal gauge [2]. Its residual pure-gauge transformations are generated by Carrollian Noether screenings, and its gauge-invariant observables are row-determinant

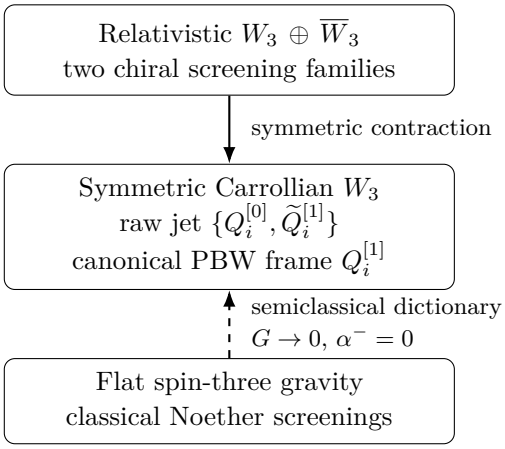


Figure 1: The parent systems contract to a raw screening jet; its fixed-target representative is the canonical deformation intertwiner $Q_i^{[1]}$. The bulk charges realize the leading and raw semiclassical coefficients.

Miura combinations, paralleling free-field variables used in higher-spin quantization [9]. Together with the analysis of Carrollian quantum states in [3], these results leave a sharply defined missing step: the simple-root quantum screening jets for the proper symmetric Carrollian construction.

We address this problem for $N = 3$, the first nonlinear case. Three features distinguish the answer from the relativistic template.

First, the two roots of the relativistic screening equation obey $\beta_+\beta_- = -4$. A regular Carrollian local limit keeps one branch with $\beta = \varepsilon\beta_1 + O(\varepsilon^3)$; the reciprocal branch diverges and fixes the contracted background charge. The remnant of the two-branch relation is therefore

$$\alpha^+\beta_1 = -2, \qquad \beta_1 = -2\mu, \qquad \alpha^+\mu = 1. \tag{1.1}$$

There is only one regular local leading screening per simple root.

Second, the unbarred and barred parent charges contract to a jet rather than to two independent screenings. The leading coefficient is an ordinary screening. The first relative coefficient compares intertwiners whose target Fock momenta follow different curves. In a common PBW-flat frame it is a deformation intertwiner,

$$Q_i^{[1]} = \widetilde{Q}_i^{[1]} - \mathsf{A}_{\mathrm{T},i} Q_i^{[0]}, \qquad \mathsf{A}_{\mathrm{T},i} = \mu\,\alpha_i\cdot q^- + \nu\,\alpha_i\cdot q^+, \qquad \nu = -\alpha^-\mu^2, \tag{1.2}$$

and obeys

$$[Q_i^{[1]}, \mathcal{U}_{3,n}^{\sigma,[0]}]_{\mathrm{PBW}} + \frac{1}{2}\,\Delta\mathcal{U}_{\mathrm{T},3,n}^{\sigma,[1]} Q_i^{[0]} = 0, \qquad i = 1,2, \quad \sigma = \pm. \tag{1.3}$$

The inhomogeneous term is the connection required to compare the moving target representations; it is not removable by a scalar choice of scheme.

Third, the same distinction has a semiclassical bulk realization. The leading bulk Noether charges match $Q_i^{[0],\mathrm{cl}}$ and the barred-linear charges match the raw moving-frame coefficient $\widetilde{Q}_i^{[1],\mathrm{cl}}$. Passing to a fixed boundary target frame produces equation (1.2). Thus the bulk does not supply an ordinary second quantum screening.

The leading theorem gives the inclusion

$$W_3^{\mathrm{Car}} \subseteq \bigcap_{i=1}^{2} \ker\left(\operatorname{ad} Q_i^{[0]}\right) \tag{1.4}$$

as an identity of intertwiners on the Fock families specified below. Equality is not proved. Here $\operatorname{ad} Q_i^{[0]}$ denotes the source–target intertwiner differential $U_{\mathrm{T}} Q_i^{[0]} - Q_i^{[0]} U_{\mathrm{S}}$, rather than an adjoint action in a single endomorphism algebra. We likewise make no claim that the two jet coefficients form a pair of ordinary quantum screenings, nor do we claim screening–screening closure, Serre relations, a BRST or Felder complex, or a quantum-group structure. The distinction between $\widetilde{Q}_i^{[1]}$ and $Q_i^{[1]}$ is maintained throughout.

The paper is organized as follows. Section 2 fixes the crossed free fields, averaged ordering and primary W_3 currents. Section 3 derives the parent screening condition, its reciprocal branch, the contracted parameter hierarchy and the leading operator theorem. Section 4 constructs the finite screening jet and exhibits the A_1 obstruction. Section 5 proves the nonlinear deformation-intertwiner equation by both finite-parent contraction and a direct averaged-order OPE. Section 6 establishes the semiclassical bulk dictionary. Technical formulas and normalization notes are collected in the appendices.

2 Symmetric Carrollian free fields and primary W_3 currents

2.1 Crossed Heisenberg algebra, potentials and ordering

Let

$$\mathfrak{h} = \left\{ x \in \mathbb{C}^3 \,\middle|\, \sum_{a=1}^{3} x_a = 0 \right\}, \qquad \kappa_{ab} = \delta_{ab} - \frac{1}{3}, \tag{2.1}$$

and choose the simple roots

$$\alpha_1 = (1,-1,0), \qquad \alpha_2 = (0,1,-1), \qquad \alpha_i \cdot \alpha_j = A_{ij}, \qquad A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{2.2}$$

The symmetric Carrollian currents $P_a = J_a^+$ and $N_a = J_a^-$ satisfy

$$P_a(z)P_b(w) \sim 0, \qquad N_a(z)N_b(w) \sim 0, \qquad N_a(z)P_b(w) \sim \frac{\kappa_{ab}/2}{(z-w)^2}, \tag{2.3}$$

with $\sum_a P_a = \sum_a N_a = 0$. Introducing $P_a = \partial\varphi_a^+$ and $N_a = \partial\varphi_a^-$ fixes

$$\varphi_a^-(z)\varphi_b^+(w) \sim \frac{\kappa_{ab}}{2}\log(z-w), \qquad [p_a^-, q_b^+] = [p_a^+, q_b^-] = \frac{\kappa_{ab}}{2}. \tag{2.4}$$

The plus sign in equation (2.4) follows from the parent convention $J_a(z)J_b(w) \sim \kappa_{ab}(z-w)^{-2}$ when $J_a = \partial\varphi_a$. Relative to a canonically normalized real boson, $\varphi = \mathrm{i}\phi_{\mathrm{real}}$; this real-form rotation is responsible for the imaginary background charge on the gravity branch.

For later use it is important that the crossed algebra is not merely the limiting algebra of the contraction. If $a_{a,n}$ and $\bar{a}_{a,n}$ are the modes of the two parent Heisenberg systems, with

$$[a_{a,m}, a_{b,n}] = m\kappa_{ab}\delta_{m+n,0}, \qquad [\bar{a}_{a,m}, \bar{a}_{b,n}] = m\kappa_{ab}\delta_{m+n,0}, \tag{2.5}$$

then the ultrarelativistic, equal-time embedding is

$$P_{a,n} = \frac{\varepsilon}{2}(a_{a,n} + \bar{a}_{a,-n}), \qquad N_{a,n} = \frac{1}{2\varepsilon}(a_{a,n} - \bar{a}_{a,-n}). \tag{2.6}$$

The reversal of the barred mode changes the sign of its central term, and direct substitution gives

$$[P_{a,m}, P_{b,n}] = [N_{a,m}, N_{b,n}] = 0, \qquad [N_{a,m}, P_{b,n}] = \frac{m}{2}\kappa_{ab}\delta_{m+n,0} \tag{2.7}$$

for every $\varepsilon \neq 0$. With $t = \varepsilon^2$, the parent Weyl embeddings are understood after scalar extension to $\mathbb{C}((t^{1/2}))$; the contracted currents and the transported screening vertices considered below lie in the regular t-adic subcategory over $\mathbb{C}[[t]]$. Thus the crossed Heisenberg brackets themselves are ε-independent; the remaining contraction data reside in the composite Miura fields, background charges and parent vertex embeddings. This exact finite-ε statement is what permits the parent screening relations to be transported before coefficient extraction; it also prevents the opposite parent polarizations from being identified prematurely.

Products containing one N-field use the symmetric ordering inherited from the two parent polarizations,

$$\mathrm{Av}(NP_1 \cdots P_r) := \frac{1}{2}\left(:NP_1 \cdots P_r: + :P_1 \cdots P_r N:\right). \tag{2.8}$$

This is not a freely chosen limiting convention: it is obtained by retaining the two parent normal-order orientations until after coefficient extraction [1].

2.2 Contracted Miura currents

For three traceless parent currents X_a, define the ordered row-determinant

$$\mathcal{R}(X, a) = \prod_{j=1}^{3}(a\partial - X_j) = (a\partial)^3 - u_2(a\partial) - u_3, \qquad w(X, a) := u_3 - \frac{a}{2}\partial u_2. \tag{2.9}$$

The symmetric contraction embeds the two parent orientations, for $t = \varepsilon^2$, as

$$X^{\mathrm{u}}_a = \varepsilon^{-1}P_a + \varepsilon N_a, \quad a_{\mathrm{u}} = \varepsilon^{-1}\alpha^+ + \varepsilon\alpha^-, \qquad X^{\mathrm{b}}_a = \varepsilon^{-1}P_a - \varepsilon N_a, \quad a_{\mathrm{b}} = \varepsilon^{-1}\alpha^+ - \varepsilon\alpha^-. \tag{2.10}$$

After the barred equal-time map,

$$\mathbb{W}^+(t) = \frac{\varepsilon^3}{2}(w_{\mathrm{u}} + w_{\mathrm{b}}), \qquad \mathbb{W}^-(t) = \frac{\varepsilon}{2}(w_{\mathrm{u}} - w_{\mathrm{b}}). \tag{2.11}$$

Parity under $(N, \alpha^-) \mapsto (-N, -\alpha^-)$ gives

$$\mathbb{W}^+(t) = \mathcal{U}_3^+ + t^2\mathcal{U}^{(2)}_{3,+}, \qquad \mathbb{W}^-(t) = \mathcal{U}_3^- + t^2\mathcal{U}^{(2)}_{3,-}. \tag{2.12}$$

The universal primaries have no coefficient of order t; their represented actions on moving Fock modules need not share this property.

In the reduced variables $P_3 = -P_1 - P_2$ and $N_3 = -N_1 - N_2$, the spin-two fields are

$$U_2^+ = P_1^2 + P_1P_2 + P_2^2 - \alpha^+(2P_1' + P_2'), \tag{2.13}$$

$$U_2^- = 2\,\mathrm{Av}(N_1P_1) + \mathrm{Av}(N_1P_2) + \mathrm{Av}(N_2P_1) + 2\,\mathrm{Av}(N_2P_2) - \alpha^+(2N_1' + N_2') - \alpha^-(2P_1' + P_2'). \tag{2.14}$$

The primary spin-three fields are

$$\begin{aligned}\mathcal{U}_3^+ =& -P_1^2P_2 - P_1P_2^2 + \alpha^+\left(\frac{3}{2}P_1P_2' + \frac{1}{2}P_1'P_2 + P_2P_2'\right) - \frac{(\alpha^+)^2}{2}P_2'', \qquad (2.15)\\ \mathcal{U}_3^- =& -2\,\mathrm{Av}(N_1P_1P_2) - \mathrm{Av}\Big(N_1P_2^2\Big) - \mathrm{Av}\Big(N_2P_1^2\Big) - 2\,\mathrm{Av}(N_2P_1P_2)\\ &+\alpha^-\left(\frac{3}{2}P_1P_2' + \frac{1}{2}P_1'P_2 + P_2P_2'\right)\\ &+\alpha^+\Big(\frac{3}{2}\,\mathrm{Av}(N_1P_2') + \frac{1}{2}\,\mathrm{Av}(N_1'P_2) + \frac{1}{2}\,\mathrm{Av}(N_2P_1') + \mathrm{Av}(N_2P_2')\\ &\qquad + \frac{3}{2}\,\mathrm{Av}(N_2'P_1) + \mathrm{Av}(N_2'P_2)\Big)\\ &-\alpha^+\alpha^-P_2'' - \frac{(\alpha^+)^2}{2}N_2''. \qquad (2.16)\end{aligned}$$

The second field is the directional derivative of the first under $(P,\alpha^+)\mapsto(P+sN,\alpha^+ + s\alpha^-)$ at $s=0$. It therefore contains at most one N-field. Together with equation (2.3), this one-N filtration bounds the number of Wick contractions in every screening OPE below.

3 Parent screenings and leading Carrollian charges

3.1 Parent screening equation and the reciprocal branch

For a parent simple-root vertex $V_i =: e^{\lambda\alpha_i\cdot\varphi}:$, its spin-two weight in the conventions of equation (2.9) is

$$h(\lambda\alpha_i) = \lambda^2 + a\lambda. \qquad (3.1)$$

Thus the parent screening locus is

$$a\lambda + \lambda^2 = 1. \qquad (3.2)$$

In the A_1 single-field normalization $\varphi_{\mathrm{sf}}^{\pm} = -\alpha_1\cdot\varphi^{\pm}/2$ and $\beta = -2\lambda$, this becomes

$$\beta^2 - 2a\beta - 4 = 0, \qquad \beta_+\beta_- = -4, \qquad \beta_+ + \beta_- = 2a. \qquad (3.3)$$

Writing $\beta_+ = 2b$ and $\beta_- = -2/b$ gives $a = b - b^{-1}$. Thus the two roots are the usual reciprocal Coulomb-gas momenta, expressed in the real form selected by equation (2.4). The product relation, rather than the individual parametrization, is the invariant datum that survives the contraction. The same locus screens the parent spin-three primary. For A_2 this follows directly from the ordered Miura OPE: the non-derivative residue factorizes by $a\lambda+\lambda^2-1$, and on equation (3.2) the singular parts are total derivatives. In reduced variables $X_3 = -X_1 - X_2$, explicit witnesses are

$$\mathrm{Sing}[u_2(z)V_i(w)] = \partial_w\left(\frac{V_i(w)}{z-w}\right), \qquad i=1,2, \qquad (3.4)$$

$$\mathrm{Sing}[u_3(z)V_1(w)] = \partial_w\left(\frac{(X_1+X_2)V_1(w)}{z-w}\right), \qquad (3.5)$$

$$\mathrm{Sing}[u_3(z)V_2(w)] = \partial_w\left(-\frac{aV_2(w)}{(z-w)^2} - \frac{X_1V_2(w)}{z-w}\right). \qquad (3.6)$$

These formulas hold on equation (3.2) and are the finite objects contracted in section 5; no higher-rank extrapolation is needed for the present theorem.

Take a regular single-field branch

$$\beta = \varepsilon\beta_1 + \varepsilon^3\beta_3 + O(\varepsilon^5), \qquad \bar{\beta} = \varepsilon\beta_1 + \varepsilon^3\bar{\beta}_3 + O(\varepsilon^5), \tag{3.7}$$

and $a = \varepsilon^{-1}\alpha^+ + \varepsilon\alpha^-$, $\bar{a} = \varepsilon^{-1}\alpha^+ - \varepsilon\alpha^-$. Expanding equation (3.3) gives

$$\alpha^+\beta_1 = -2, \qquad \alpha^+\beta_3 = \frac{1}{2}\beta_1^2 - \alpha^-\beta_1, \qquad \alpha^+\bar{\beta}_3 = \frac{1}{2}\beta_1^2 + \alpha^-\beta_1. \tag{3.8}$$

Equivalently, in the root normalization $\beta_1 = -2\mu$,

$$\alpha^+\mu = 1, \qquad \mu_3 = -\mu^3 - \alpha^-\mu^2, \qquad \bar{\mu}_3 = -\mu^3 + \alpha^-\mu^2, \qquad \nu := \frac{\mu_3 - \bar{\mu}_3}{2} = -\alpha^-\mu^2. \tag{3.9}$$

The leading equation can also be obtained without referring back to the parent quadratic. In the A_1 normalization $[J_m^+, J_n^-] = m\delta_{m+n,0}/4$, set

$$M = 2 : J^+J^+ : +2\alpha^+\partial J^+, \qquad L = 4\,\mathrm{Av}\Big(J^+J^-\Big) + 2\alpha^+\partial J^- + 2\alpha^-\partial J^+. \tag{3.10}$$

For $E =: e^{\beta_1\varphi^+} :$, the plus field has no contraction with the exponent, and a single crossed contraction gives

$$M(z)E(w) \sim 0, \qquad L(z)E(w) \sim -\frac{\alpha^+\beta_1}{2}\frac{E(w)}{(z-w)^2} + \frac{\partial E(w)}{z-w}. \tag{3.11}$$

Consequently the weight-one condition is again $\alpha^+\beta_1 = -2$. The parent expansion and the contracted OPE are independent derivations: the former retains the reciprocal branch, whereas the latter shows directly that the leading relation is an exact quantum condition and does not depend on the averaged ordering.

The reciprocal parent solution $-4/\beta$ diverges as ε^{-1}, and

$$\alpha^+ = \lim_{\varepsilon\to 0}\varepsilon a = -\frac{2}{\beta_1}. \tag{3.12}$$

Hence the second parent branch does not give a second regular local leading density; its information is retained in the background charge.

For A_2 the counting is therefore

$$2 \text{ simple roots} \times 1 \text{ regular branch} \times 2 \text{ jet coefficients} = 4, \tag{3.13}$$

matching the two exponential and two barred-linear elementary bulk charges encountered in section 6. On the gravity branch $c_L = 0$, $c_M \neq 0$ implies $\alpha^- = 0$ and $\alpha^+ \neq 0$; then $\nu = 0$, so the strict tangent density is purely $: (\mu\alpha_i \cdot \varphi^-)E_i :$, exactly the field content required by the barred-linear bulk charges. We keep α^- arbitrary until the bulk comparison because the quantum Carrollian algebra and the deformation-intertwiner theorem exist away from this locus.

3.2 Carrier, non-vanishing and the leading theorem

For each simple root define

$$E_i(z) =: \exp\big(\mu\alpha_i \cdot \varphi^+(z)\big) : \ . \tag{3.14}$$

Because $\varphi^+\varphi^+ \sim 0$, the leading exponentials have no oscillator self-contractions. Let $\mathcal{P}_{\mathrm{fin}}$ be the finite PBW algebra of crossed modes and $\widehat{\mathcal{P}}_E$ its energy completion. On the lattice

$$\Lambda_0 = \bigoplus_{i=1}^{2} \mathbb{Z}\,\frac{\mu}{2}\kappa\alpha_i, \qquad \mathcal{D}^{(0)} = \mathcal{P}_{\mathrm{fin}} \otimes \mathbb{C}[\Lambda_0], \tag{3.15}$$

The leading fields act on the coset sum

$$\mathbb{F}_{\lambda_0} := \bigoplus_{\eta\in\Lambda_0} \widehat{\mathcal{F}}(\lambda_0^+, \lambda_0^- + \eta). \tag{3.16}$$

Here $\mathcal{P}_{\mathrm{fin}}$ consists of finite crossed-mode PBW monomials, while completion is taken degree by degree only in the target. The absence of a $\varphi^+\varphi^+$ contraction permits the trivial lattice cocycle for the leading exponentials; the screening operators are therefore intertwiners between summands of equation (3.16), rather than endomorphisms of a single Fock module. The half-shift follows from equation (2.4),

$$p^- e^{\mu\alpha_i\cdot q^+} = e^{\mu\alpha_i\cdot q^+}\left(p^- + \frac{\mu}{2}\kappa\alpha_i\right). \tag{3.17}$$

For integral monodromy, the residue

$$Q_i^{[0]} = 2\pi\mathrm{i}\operatorname*{Res}_{z=0} E_i(z) : \mathcal{F}(\lambda^+, \lambda^-) \longrightarrow \widehat{\mathcal{F}}\left(\lambda^+, \lambda^- + \frac{\mu}{2}\kappa\alpha_i\right) \tag{3.18}$$

is continuous in the target energy topology. It is non-zero: choose a resonant vacuum with $\mu\alpha_i \cdot \lambda_0^+ = -1$; then

$$E_i(z)|\lambda_0\rangle = z^{-1} e^{\mu\alpha_i\cdot q^+}|\lambda_0\rangle + \cdots, \tag{3.19}$$

whose vacuum-to-vacuum residue is non-vanishing.

The spin-two OPE fixes equation (1.1). For the primary spin-three fields, $\mathcal{U}_3^+(z)E_i(w)$ is regular and

$$\mathrm{Sing}[\mathcal{U}_3^-(z)E_1(w)] = \partial_w\left[\left(\frac{P_1+P_2}{2(z-w)} + \frac{1}{4\mu(z-w)^2}\right)E_1(w)\right], \tag{3.20}$$

$$\mathrm{Sing}[\mathcal{U}_3^-(z)E_2(w)] = \partial_w\left[\left(-\frac{P_1}{2(z-w)} - \frac{1}{4\mu(z-w)^2}\right)E_2(w)\right]. \tag{3.21}$$

On the horizontal periodic sectors the integrated boundaries vanish.

Theorem 3.1 (Leading simple-root screenings)**.** *On $\alpha^+\mu = 1$, for $i = 1, 2$, $\sigma = \pm$ and all $n \in \mathbb{Z}$,*

$$[Q_i^{[0]}, U_{2,n}^{\sigma}] = 0, \qquad [Q_i^{[0]}, \mathcal{U}_{3,n}^{\sigma}] = 0. \tag{3.22}$$

The brackets are identities of intertwiners between the momentum sectors in equation (3.18).

4 The finite screening jet and the A_1 obstruction

4.1 Exact branches, affine vertices and horizontal families

Set $t = \varepsilon^2$. The two regular root branches are the unique formal solutions

$$t\mu_{\mathrm{u}}(t)^2 + (\alpha^+ + t\alpha^-)\mu_{\mathrm{u}}(t) = 1, \qquad t\mu_{\mathrm{b}}(t)^2 + (\alpha^+ - t\alpha^-)\mu_{\mathrm{b}}(t) = 1, \tag{4.1}$$

with common constant term $\mu = 1/\alpha^+$. The transported one-forms contain affine factors,

$$V_{i,\mathrm{u}}(z,t) = z^{t\mu_{\mathrm{u}}(t)^2} :\exp\Big[\mu_{\mathrm{u}}(t)\alpha_i \cdot (\varphi^+ + t\varphi^-)\Big]:_{\mathrm{n}}, \tag{4.2}$$

$$V_{i,\mathrm{b}}(z,t) = z^{t\mu_{\mathrm{b}}(t)^2} :\exp\Big[\mu_{\mathrm{b}}(t)\alpha_i \cdot (\varphi^+ - t\varphi^-)\Big]:_{\mathrm{an}}\,. \tag{4.3}$$

The second line is anti-normally ordered in the common equal-time polarization. Its conversion to the common normal order, with radial regulator $\rho < 1$, is multiplication by

$$\bar{Z}_{t,\rho} = (1-\rho)^{-2t\mu_{\mathrm{b}}(t)^2}, \tag{4.4}$$

a scalar independent of the insertion point. A finite subtraction changes the first coefficient only by a multiple of $Q_i^{[0]}$.

For an integer monodromy label m, write $\lambda_i^{\pm}(t) = \alpha_i \cdot \lambda^{\pm}(t)$. The simultaneous horizontal section is

$$\lambda_i^+(t) = \frac{1}{2}\left(\frac{m - t\mu_{\mathrm{u}}(t)^2}{\mu_{\mathrm{u}}(t)} + \frac{m - t\mu_{\mathrm{b}}(t)^2}{\mu_{\mathrm{b}}(t)}\right), \tag{4.5}$$

$$t\lambda_i^-(t) = \frac{1}{2}\left(\frac{m - t\mu_{\mathrm{u}}(t)^2}{\mu_{\mathrm{u}}(t)} - \frac{m - t\mu_{\mathrm{b}}(t)^2}{\mu_{\mathrm{b}}(t)}\right). \tag{4.6}$$

Both branch monodromies vanish identically as formal series; the apparent factor t^{-1} in equation (4.6) is removable.

Let $R = \mathbb{C}[[t]]$ and identify the moving source and target Fock families by the PBW-flat map $X|\lambda(t)\rangle \mapsto X|\lambda(0)\rangle$. After the scalar conversion equation (4.4), the two parent charges lie in a common topologically free Hom-space. More explicitly, the source and target trivializations give a fixed module

$$\mathcal{H}_i = \mathrm{Hom}_R^{\mathrm{cont}}\left(\mathcal{F}_{\mathrm{S},i}(0)[[t]], \widehat{\mathcal{F}}_{\mathrm{T},i}(0)[[t]]\right), \tag{4.7}$$

and preserve every negative-mode PBW monomial together with a chosen normalized vacuum. The exact horizontality equations determine the moving zero-mode eigenvalues; they are not discarded by the trivialization. This distinction is essential: PBW-flatness identifies basis vectors, whereas the target action of a current still differentiates along the target momentum curve.

Writing them as $\widehat{Q}_{i,\mathrm{u}}(t)$ and $\widehat{Q}_{i,\mathrm{b}}(t)$, define

$$Q_i^{(0)}(t) = \frac{\widehat{Q}_{i,\mathrm{u}}(t) + \widehat{Q}_{i,\mathrm{b}}(t)}{2}, \qquad \widetilde{Q}_i^{(1)}(t) = \frac{\widehat{Q}_{i,\mathrm{u}}(t) - \widehat{Q}_{i,\mathrm{b}}(t)}{2t}. \tag{4.8}$$

Their constant coefficients are $Q_i^{[0]}$ and $\widetilde{Q}_i^{[1]}$. The numerator in the second expression is divisible by t in the common Hom-space. The first coefficient is finite and non-zero. Indeed, in the A_1 reduction choose the horizontal resonant vacuum with monodromy label $m = -2$. The level-one projection of the half-difference contains

$$\pi_{J_{-1}^-}\widetilde{Q}^{[1]}|\lambda_{-2}(0)\rangle \;\propto\; \beta_1 J_{-1}^- e^{\beta_1 q^+}|\lambda_{-2}(0)\rangle \neq 0, \tag{4.9}$$

whereas $Q^{[0]}$ has no J_{-1}^- component. This proves both non-vanishing and linear independence from the leading charge. Notice that the level-one argument applies here, to the first

relative coefficient; the non-vanishing of $Q_i^{[0]}$ was instead proved by the resonant vacuum residue in equation (3.19).

The local density separates into strict odd and common even parts,

$$\mathcal{S}^{[1]}_{i,\mathrm{odd}} =: (\mu\alpha_i\cdot\varphi^- + \nu\alpha_i\cdot\varphi^+)E_i :, \tag{4.10}$$

$$\mathcal{S}^{[1]}_{i,\mathrm{even}} =: \left(-\mu^3\alpha_i\cdot\varphi^+ + \mu^2\log\frac{w}{w_0}\right)E_i : . \tag{4.11}$$

The logarithm comes from the affine one-form and is interpreted by the based residue in section A.

4.2 Local A_1 obstruction and spin-two connection

The strict odd coefficient cannot be treated as an ordinary local screening. In the A_1 single-field normalization let $S_{A,B,C} =: (A\varphi^- + B\varphi^+ + C)E :$, with $E =: e^{\beta_1\varphi^+} :$. A complete crossed-Wick calculation gives the anomalous double-pole coefficient

$$K(A,B) = \frac{A\beta_1}{4} - \frac{\alpha^+B}{2} - \frac{\alpha^-A}{2}. \tag{4.12}$$

For the strict odd values $A=\beta_1$ and $B=-(\alpha^-\beta_1)/\alpha^+$, the screening locus gives

$$K = \frac{\beta_1^2}{4} \neq 0. \tag{4.13}$$

The even parameter/affine tangent cancels this local A_1 diagnostic in the full coefficient, but the local calculation alone neither defines the logarithmic contour nor identifies the two moving targets. Those data are supplied by the exact construction above. In particular, the W_3 averaged-order mechanism of section 5 must not be replaced by an odd/even remainder cancellation.

The two target curves have relative tangent, in the single-field normalization,

$$\Delta p^+ = \frac{\beta_1}{2}, \qquad \Delta p^- = \frac{B_{\mathrm{diff}}}{2}, \qquad B_{\mathrm{diff}} = -\frac{\alpha^-\beta_1}{\alpha^+}. \tag{4.14}$$

With the target-first bracket

$$[Q,U]_{\mathrm{PBW}} := U_{\mathrm{T}}^{[0]}Q - QU_{\mathrm{S}}^{[0]}, \tag{4.15}$$

coefficient extraction from the exact parent intertwiners yields

$$[Q^{[1]},U_n^{[0]}]_{\mathrm{PBW}} + \frac{1}{2}\Delta U^{[1]}_{\mathrm{T},n}Q^{[0]} = 0, \qquad U = M, L, \tag{4.16}$$

where

$$\Delta M^{[1]}_{\mathrm{T},n} = \frac{\beta_1}{2}(4J_n^+ - 2\alpha^+\delta_{n0}), \tag{4.17}$$

$$\Delta L^{[1]}_{\mathrm{T},n} = \frac{\beta_1}{2}(4J_n^- - 2\alpha^-\delta_{n0}) + \frac{B_{\mathrm{diff}}}{2}(4J_n^+ - 2\alpha^+\delta_{n0}). \tag{4.18}$$

On a horizontal sector with J_0^+-eigenvalue m/β_1, the zero-mode equations reduce to

$$[Q^{[1]},M_0]_{\mathrm{PBW}} = -(m+1)Q^{[0]}, \qquad [Q^{[1]},L_0]_{\mathrm{PBW}} = -\frac{\beta_1^2}{4}Q^{[0]}. \tag{4.19}$$

The magnitude of the second inhomogeneity is exactly the local Wick obstruction $K = \beta_1^2/4$ of equation (4.13); in the target-first convention it enters as $-KQ^{[0]}$, and is therefore reinterpreted as the target-connection term rather than as a failure of the exact parent family. A scalar change $Q^{[1]} \mapsto Q^{[1]} + cQ^{[0]}$ cannot remove it because the leading charge commutes with L_0. Thus the Virasoro sector already requires a deformation intertwiner.

5 The nonlinear W_3 deformation intertwiner

5.1 Relative target motion and nonlinear connection

For the simple root α_i, the relative first tangent of the two target curves is

$$\Delta p_i^+ = \mu\alpha_i, \qquad \Delta p_i^- = \nu\alpha_i. \tag{5.1}$$

At the common leading target, let $\mathfrak{d}_i^+$ and $\mathfrak{d}_i^-$ denote the zero-mode shift derivatives in the P and N directions. Define

$$\mathcal{A}_i := \mathfrak{d}_i^+ \mathcal{U}_3^+, \qquad \mathcal{B}_i := \mathfrak{d}_i^+ \mathcal{U}_3^-. \tag{5.2}$$

The directional-Miura relation between equations (2.15) and (2.16) implies

$$\mathfrak{d}_i^- \mathcal{U}_3^+ = 0, \qquad \mathfrak{d}_i^- \mathcal{U}_3^- = \mathcal{A}_i. \tag{5.3}$$

Consequently the represented target-action differences are

$$\Delta\mathcal{U}_{\mathrm{T},3}^{+,[1]} = \mu\mathcal{A}_i, \qquad \Delta\mathcal{U}_{\mathrm{T},3}^{-,[1]} = \mu\mathcal{B}_i + \nu\mathcal{A}_i. \tag{5.4}$$

The explicit fields are recorded in section B.

5.2 Finite-parent averaged ordering

For each parent orientation $r \in \{\mathrm{u}, \mathrm{b}\}$, the exact screening charge is an intertwiner before contraction:

$$U_{\mathrm{T},r}(t)\widehat{Q}_{i,r}(t) - \widehat{Q}_{i,r}(t)U_{\mathrm{S},r}(t) = 0. \tag{5.5}$$

The two identities live over different target curves, so their Taylor coefficients may be subtracted only after the PBW identifications leading to equation (4.7). In the branch-adapted moving frames the constant term of their half-difference is homogeneous. In a fixed target frame, differentiating the target trivialization contributes one half of the relative target-action tangent. This is the origin of the factor $1/2$ in equations (5.4) and (5.18); it is fixed by the symmetric half-difference and is not a normalization choice.

The direct tangent OPE must be extracted before the two parent normal-order orientations are collapsed. For a parent vertex $V_\lambda =: e^{\lambda\alpha_i\cdot\varphi} :$, the parent primary satisfies

$$\mathrm{Sing}[w(z)V_\lambda(w)] = \partial_w\left[\left(\frac{B_{i,1}}{z-w} + \frac{B_{i,2}}{(z-w)^2}\right)V_\lambda(w)\right] \tag{5.6}$$

precisely on $a\lambda + \lambda^2 = 1$. Substitution of the two embeddings equation (2.10) and branches equation (4.1), followed only then by coefficient extraction, produces the averaged order equation (2.8). Since the parent singular OPE contains finitely many poles and contractions, this extraction commutes with the singular-part map.

For the strict odd density equation (4.10), the plus-primary OPEs are

$$\mathrm{Sing}[\mathcal{U}_3^+(z)\mathcal{S}^{[1]}_{1,\mathrm{odd}}(w)] = \partial_w\left[\left(\frac{P_1+P_2}{2(z-w)}+\frac{1}{4\mu(z-w)^2}\right)E_1\right], \tag{5.7}$$

$$\mathrm{Sing}[\mathcal{U}_3^+(z)\mathcal{S}^{[1]}_{2,\mathrm{odd}}(w)] = \partial_w\left[\left(-\frac{P_1}{2(z-w)}-\frac{1}{4\mu(z-w)^2}\right)E_2\right]. \tag{5.8}$$

For the minus-primary, put

$$H_i = \alpha_i\cdot\varphi^- - \alpha^-\mu\,\alpha_i\cdot\varphi^+. \tag{5.9}$$

The complete averaged-order result is

$$\mathrm{Sing}[\mathcal{U}_3^-(z)\mathcal{S}^{[1]}_{i,\mathrm{odd}}(w)]_{\mathrm{Av}} = \partial_w\left[\left(\frac{\Xi_{i,1}}{z-w}+\frac{\Xi_{i,2}}{(z-w)^2}\right)E_i(w)\right], \tag{5.10}$$

with

$$\Xi_{1,2} = \frac{H_1+\alpha^-}{4}, \qquad \Xi_{1,1} = \frac{N_1+N_2+\mu H_1(P_1+P_2)}{2}, \tag{5.11}$$

$$\Xi_{2,2} = -\frac{H_2+\alpha^-}{4}, \qquad \Xi_{2,1} = -\frac{N_1+\mu H_2 P_1}{2}. \tag{5.12}$$

No remainder survives.

The finite-parent average supplies the contact omitted by a premature commutative replacement,

$$\begin{aligned} i=1: &\quad (\delta\mathcal{O}_{1,2},\delta\mathcal{O}_{1,1}) = \left(\frac{\mu}{4w},\frac{\mu^2(P_1+P_2)}{2w}\right),\\ i=2: &\quad (\delta\mathcal{O}_{2,2},\delta\mathcal{O}_{2,1}) = \left(-\frac{\mu}{4w},-\frac{\mu^2 P_1}{2w}\right). \end{aligned} \tag{5.13}$$

and $\partial_w(w^{-1}) = -w^{-2}$ supplies the associated second-order terms. The contact is fixed by the two parent orientations; it is not an adjustable counterterm. After it is restored, the strict odd OPE is separately exact. The common even/affine coefficient equation (4.11) is also separately exact, with witnesses

$$(\Xi^{\mathrm{even}}_{1,1},\Xi^{\mathrm{even}}_{1,2}) = \left(\frac{\mu^2K_1(P_1+P_2)}{2},\frac{\mu K_1}{4}\right), \qquad (\Xi^{\mathrm{even}}_{2,1},\Xi^{\mathrm{even}}_{2,2}) = \left(-\frac{\mu^2K_2P_1}{2},-\frac{\mu K_2}{4}\right), \tag{5.14}$$

where $K_i = \log(w/w_0) - \mu\alpha_i\cdot\varphi^+$. Thus the corrected mechanism is not a cancellation of an odd remainder by the even channel: both coefficients are separately total derivatives after the inherited averaged ordering is retained.

5.3 Horizontal boundary and canonical target frame

A logarithmic total derivative cannot be discarded before imposing monodromy. For a weight-one field with residual exponent a, the based residue obeys

$$\operatorname*{LRes}_{w_0}\partial_w\big(w^{n+1}V(w)\big) = \frac{e^{2\pi\mathrm{i}a}-1}{2\pi\mathrm{i}}\,w_0^{n+1}V(w_0). \tag{5.15}$$

The exact sections equations (4.5) and (4.6) set the residual exponents of both parent branches to zero as formal series. Their boundaries therefore vanish before coefficient extraction, giving homogeneous raw moving-frame closure,

$$[\widetilde{Q}_i^{[1]}, \mathcal{U}_{3,n}^{\sigma,[0]}]_{\text{moving}} = 0. \tag{5.16}$$

In a fixed target frame, subtract the target-frame generator as in equation (1.2). The crossed zero-mode brackets give

$$[\mathcal{U}_3^+, \mathsf{A}_{\mathrm{T},i}] = \frac{\mu}{2}\mathcal{A}_i, \qquad [\mathcal{U}_3^-, \mathsf{A}_{\mathrm{T},i}] = \frac{1}{2}(\mu\mathcal{B}_i + \nu\mathcal{A}_i). \tag{5.17}$$

Combining equations (5.16) and (5.17) proves the nonlinear theorem.

Theorem 5.1 (Deformation-intertwiner equation)**.** *For each simple root $i = 1, 2$, every $n \in \mathbb{Z}$ and $\sigma = \pm$,*

$$[Q_i^{[1]}, \mathcal{U}_{3,n}^{\sigma,[0]}]_{\mathrm{PBW}} + \frac{1}{2}\Delta\mathcal{U}_{\mathrm{T},3,n}^{\sigma,[1]} Q_i^{[0]} = 0. \tag{5.18}$$

Equivalently,

$$[Q_i^{[1]}, \mathcal{U}_{3,n}^{+,[0]}]_{\mathrm{PBW}} = -\frac{\mu}{2}(\mathcal{A}_i)_n Q_i^{[0]}, \tag{5.19}$$

$$[Q_i^{[1]}, \mathcal{U}_{3,n}^{-,[0]}]_{\mathrm{PBW}} = -\frac{1}{2}(\mu\mathcal{B}_i + \nu\mathcal{A}_i)_n Q_i^{[0]}. \tag{5.20}$$

The operator $Q_i^{[1]}$ is finite and non-zero, but it is not an ordinary second screening of the fixed Carrollian algebra.

A change of base point, regulator subtraction or normalized vacuum may shift $Q_i^{[1]} \mapsto Q_i^{[1]} + c_i Q_i^{[0]}$. Because theorem 3.1 holds, this triangular freedom leaves equation (5.18) invariant. It exhausts the scalar scheme ambiguity; it does not remove the nonlinear target connection.

6 Semiclassical bulk realization

6.1 Normalization triangle and the $N = 2$ anchor

For the flat higher-spin theory of [2],

$$k_{\mathrm{CS}} = \frac{1}{8\pi G}, \qquad \tau_s = \frac{12}{s(s^2-1)}, \qquad a_s = \sqrt{\pi k_{\mathrm{CS}}\tau_s}, \qquad a_s^2 = \frac{3}{2Gs(s^2-1)}. \tag{6.1}$$

The symmetric Carrollian central charges are

$$c_L = -4s(s^2-1)\alpha^+\alpha^-, \qquad c_M = -2s(s^2-1)(\alpha^+)^2. \tag{6.2}$$

The gravity branch is therefore

$$\alpha^- = 0, \qquad \alpha^+ = \mathrm{i}a_s, \qquad c_L = 0, \qquad c_M = \frac{3}{G}, \qquad \mu = (\alpha^+)^{-1} = -\frac{\mathrm{i}}{a_s} = O(\sqrt{G}). \tag{6.3}$$

In particular,

$$a_2 = \frac{1}{2\sqrt{G}}, \qquad \alpha_{s=2}^+ = \frac{\mathrm{i}}{2\sqrt{G}}. \tag{6.4}$$

At $s = 2$, the complete coefficient solve starts from $J^+ = A\psi'$, $J^- = B\bar{\psi}' + C\psi'$ and $\alpha^+ = Da_2$, together with the most general triangular linear relation between the boundary and bulk spin-two fields. The central charge, the two Miura polynomials and the crossed bracket admit only two globally conjugate solutions. Choosing the branch in equation (6.3) gives

$$A = B = \frac{\mathrm{i}a_2}{2}, \qquad C = 0, \qquad D = \mathrm{i}, \qquad M_{\mathrm{bdry}} = 2a_2^2 M_{\mathrm{bulk}}, \qquad L_{\mathrm{bdry}} = 2a_2^2 L_{\mathrm{bulk}}. \tag{6.5}$$

The equal signs of A and B are fixed by the real Fredenhagen–Hörl cylinder expansion $J_{\mathrm{FH}}(\phi) = \mathrm{i}\sum_n J_n e^{\mathrm{i}n\phi}$; reversing only the barred sign produces the wrong crossed mode bracket. In diagonal variables the common map is

$$\mathcal{P}_j = -a_s(D_0)_{jj}, \qquad \mathcal{N}_j = -a_s(D_1)_{jj}, \qquad P_j^{\mathrm{FH}} = -\mathrm{i}a_s(D_0)_{jj}, \qquad N_j^{\mathrm{FH}} = -\mathrm{i}a_s(D_1)_{jj}. \tag{6.6}$$

For $s = 2$, $D_0 = \psi' J_0$ and $D_1 = \bar{\psi}' J_0$; since $P_0 = \theta J_0$, the connection contains $\theta D_1 = \bar{\psi}' P_0$. This distinction fixes the notation used in the reduction.

The primitive map is

$$\mu\Phi^+ = -\psi, \qquad \mu\Phi^- = -\bar{\psi}, \tag{6.7}$$

so $E^{\mathrm{cl}} = e^{-\psi}$ and $\mathcal{S}_{\mathrm{odd}}^{[1],\mathrm{cl}} = -\bar{\psi}e^{-\psi}$, precisely the two elementary $s = 2$ bulk screening densities. With the normalization of [2], the corresponding charges are

$$S_{\kappa_4} = \frac{\kappa_4}{4\pi G}\int \mathrm{d}\phi\, e^{-\psi}, \qquad S_{\kappa_2} = -\frac{\kappa_2}{4\pi G}\int \mathrm{d}\phi\, \bar{\psi}e^{-\psi}, \tag{6.8}$$

and hence, for unit circle-integral normalization,

$$Q^{[0],\mathrm{cl}} = \frac{4\pi G}{\kappa_4} S_{\kappa_4}, \qquad \widetilde{Q}^{[1],\mathrm{cl}} = \frac{4\pi G}{\kappa_2} S_{\kappa_2}. \tag{6.9}$$

Their Hamiltonian variations annihilate both fields in equation (D.1). Thus the $s = 2$ anchor fixes not only the current normalization but also the relative sign and the leading/raw assignment of the two bulk charges.

6.2 A_2 root variables and Miura matching

For $s = 3$, choose the vector-representation weights

$$\epsilon_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}\right), \quad \epsilon_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}\right), \quad \epsilon_3 = \left(0, -\frac{2}{\sqrt{6}}\right), \tag{6.10}$$

with $\epsilon_i \cdot \epsilon_j = \delta_{ij} - 1/3$ and $\sum_i \epsilon_i \epsilon_i^{\mathrm{T}} = I_2$. The explicit diagonal field and the general weight-space parametrization fix one scalar-to-weight map in both sectors. Define

$$\chi_1 = \psi_{(L)} + 2\psi_{(W)}, \qquad \chi_2 = \psi_{(L)} - 2\psi_{(W)}, \tag{6.11}$$

and similarly $\bar{\chi}_i$. The explicit $s = 3$ bulk symplectic brackets combine into

$$\{\chi_i'(\phi), \bar{\chi}_j(\phi')\} = 16\pi G A_{ij}\delta(\phi - \phi'). \tag{6.12}$$

In particular, if

$$R = \begin{pmatrix} 1 & 2 \\ 1 & -2 \end{pmatrix}, \qquad R\begin{pmatrix} 8 & 0 \\ 0 & 6 \end{pmatrix} R^{\mathrm{T}} = \begin{pmatrix} 32 & -16 \\ -16 & 32 \end{pmatrix} = 16A, \tag{6.13}$$

then the two rows of R are exactly the transformations from the (L, W) scalar directions to (χ_1, χ_2). The identity equation (6.12) is therefore a direct matrix relation, not a root-dependent rescaling. Thus the distinct $8\pi G$ and $6\pi G$ coefficients of the (L, W) scalar directions are the components of one Cartan-matrix-valued bracket, not separate root normalizations.

The primitive dictionary is forced to be

$$\mu\Phi_i^+ = -\chi_i, \qquad \mu\Phi_i^- = -\bar{\chi}_i. \tag{6.14}$$

Differentiation and tracelessness give

$$P_1 = -\frac{2\chi_1' + \chi_2'}{3\mu}, \quad P_2 = \frac{\chi_1' - \chi_2'}{3\mu}, \qquad N_1 = -\frac{2\bar{\chi}_1' + \bar{\chi}_2'}{3\mu}, \quad N_2 = \frac{\bar{\chi}_1' - \bar{\chi}_2'}{3\mu}. \tag{6.15}$$

Substitution into equations (2.13) to (2.16) yields four exact relations,

$$U_2^+ = -\frac{4}{\mu^2}M_{(L)}, \qquad U_2^- = -\frac{4}{\mu^2}L_{(L)}, \qquad \mathcal{U}_3^+ = -\frac{8}{\mu^3}M_{(W)}, \qquad \mathcal{U}_3^- = -\frac{8}{\mu^3}L_{(W)}. \tag{6.16}$$

There is no root-dependent normalization and no spin-mixing improvement. The explicit bulk polynomials and the $N = 2$ coefficient equations are displayed in section D.

6.3 Bulk screenings and the raw coefficient

The four simple-root bulk charges are

$$S_{8,i} = \frac{\kappa_{8,i}}{8\pi G}\int \mathrm{d}\phi\, e^{-\chi_i}, \qquad S_{2,i} = -\frac{\kappa_{2,i}}{8\pi G}\int \mathrm{d}\phi\, \bar{\chi}_i e^{-\chi_i}, \tag{6.17}$$

where $(\kappa_{8,1}, \kappa_{8,2}) = (\kappa_8, \kappa_9)$ and $(\kappa_{2,1}, \kappa_{2,2}) = (\kappa_2, \kappa_3)$. From equation (6.12),

$$\delta_{8,i}\chi_j' = 0, \qquad \delta_{8,i}\bar{\chi}_j' = 2\kappa_{8,i}A_{ij}e^{-\chi_i}, \tag{6.18}$$

$$\delta_{2,i}\chi_j' = 2\kappa_{2,i}A_{ij}e^{-\chi_i}, \qquad \delta_{2,i}\bar{\chi}_j' = -2\kappa_{2,i}A_{ij}\bar{\chi}_i e^{-\chi_i}. \tag{6.19}$$

These transformations annihilate all four Miura observables in equation (6.16). With unit circle-integral normalization,

$$Q_i^{[0],\mathrm{cl}} = \frac{8\pi G}{\kappa_{8,i}}S_{8,i}, \qquad \widetilde{Q}_i^{[1],\mathrm{cl}} = \frac{8\pi G}{\kappa_{2,i}}S_{2,i}. \tag{6.20}$$

At the gravity point $\nu = 0$. Moreover,

$$\mathcal{S}_{i,\mathrm{odd}}^{[1]} = O(G^0), \qquad \mathcal{S}_{i,\mathrm{even}}^{[1]} = O(G), \qquad \log \bar{Z}_{t,\rho} = O(tG). \tag{6.21}$$

Thus the even/affine channel has no independent leading classical bulk charge. The bulk Noether charges act homogeneously on a single classical phase space, so the barred-linear charge realizes the raw coefficient $\widetilde{Q}_i^{[1],\mathrm{cl}}$. Its base-point dependence shifts it by a constant multiple of the leading charge, matching the triangular freedom of the boundary jet. The canonical fixed-target object remains

$$Q_i^{[1]} = \widetilde{Q}_i^{[1]} - \mathsf{A}_{\mathrm{T},i}Q_i^{[0]}. \tag{6.22}$$

Indeed, at $\nu = 0$ one may write

$$\Phi_i^-(z) = q_i^- + p_i^- \log z + \Phi_{i,\mathrm{osc}}^-(z). \tag{6.23}$$

The bulk field $\bar{\chi}_i$ contains the image of the constant mode unless a base point is chosen. Shifting that base point adds a constant to $\bar{\chi}_i$ and therefore changes the barred-linear charge by a scalar multiple of $\int e^{-\chi_i}$, precisely $\widetilde{Q}_i^{[1]} \mapsto \widetilde{Q}_i^{[1]} + c_i Q_i^{[0]}$. The fixed-target PBW subtraction removes the frame motion, not this triangular choice. The nonlocal composite bulk charges depend explicitly on a base point. No base-point-independent quantum screening–screening relation is inferred from them.

7 Discussion

The symmetric Carrollian W_3 screening problem has two distinct operator layers. The leading simple-root charges are ordinary quantum screenings: they are finite, non-zero Fock intertwiners and commute with every spin-two and primary spin-three mode on $\alpha^+\mu = 1$. Their momentum is the regular remnant of the relativistic two-branch screening equation; the reciprocal branch becomes the Carrollian background charge.

The first relative coefficients are finite and non-zero but have a different meaning. The two parent intertwiners have the same leading target and different first target tangents. Their raw coefficient is horizontal between moving fibres. A fixed-fibre comparison introduces the connection equation (5.4), and the canonical coefficient obeys the inhomogeneous relation equation (5.18). The direct OPE confirms this structure: both parent orientations must be retained until coefficient extraction, their average fixes the contact equation (5.13), and the strict odd and common even sectors are then separately total derivatives. No ad hoc contact or ordering counterterm is required.

Geometrically, the two coefficients form the flag

$$\langle Q_i^{[0]} \rangle \subset \langle Q_i^{[0]}, Q_i^{[1]} \rangle \subset \mathcal{H}_i / t^2 \mathcal{H}_i, \tag{7.1}$$

the first jet of the exact parent intertwiner family. This formulation makes the frame issue transparent. Under first-order changes of source and target trivializations with generators B_{S} and B_{T}, the coefficient transforms as

$$Q_i^{[1]} \longmapsto Q_i^{[1]} + B_{\mathrm{T}} Q_i^{[0]} - Q_i^{[0]} B_{\mathrm{S}}. \tag{7.2}$$

The canonical PBW choice sets the relative source connection to zero and identifies the target generator with $\mathsf{A}_{\mathrm{T},i}$. Scalar changes reduce to $Q_i^{[1]} \mapsto Q_i^{[1]} + c_i Q_i^{[0]}$, while the non-scalar target derivative remains as the inhomogeneous term in equation (5.18). The tangent coefficient is therefore analogous to a Jacobi field along a family of intertwiners: it is horizontal only when the moving target fibre is included in the comparison. Asking it to commute inside one fixed fibre omits precisely that geometric datum.

The bulk dictionary gives a physical interpretation. On $\alpha^- = 0$ the classical leading and barred-linear Noether screenings are the images of $Q_i^{[0],\mathrm{cl}}$ and $\widetilde{Q}_i^{[1],\mathrm{cl}}$. The PBW subtraction converts the latter into the canonical deformation intertwiner. In this precise sense the contraction produces a screening jet over representation space, not a second screening within one fixed algebra.

The proven statement remains the inclusion equation (1.4). Establishing equality would require a separate kernel or character theorem. The algebra of the screening charges, nonlocal composites, non-regular zero-mode sectors and a general-W_N construction likewise remain open. The logarithmic fields in the tangent layer suggest that generalized-module methods [13–15] may be appropriate, but no representation-theoretic closure is asserted here.

A Plane–cylinder bridge and based logarithmic residue

Use

$$w = \tau - \mathrm{i}\theta, \qquad z = e^{w}, \qquad \bar{w} = \tau + \mathrm{i}\theta, \qquad \bar{z} = e^{\bar{w}}. \tag{A.1}$$

At equal time $\bar{z} = z^{-1}$. For a Feigin–Fuchs current,

$$\mathcal{J}_w = z j_{\mathrm{pl}}(z) - \frac{\alpha}{2}, \qquad \mathcal{J}_n = a_n - \frac{\alpha}{2}\delta_{n0}, \qquad J_{\mathrm{FH}} = \mathrm{i}\mathcal{J}_w. \tag{A.2}$$

Integrating the affine current gives the potential maps

$$\varphi_{\mathrm{cyl}}(w) = -\varphi_{\mathrm{pl}}(e^{w}) + \frac{\alpha_{\mathrm{pl}}}{2} w, \qquad \bar{\varphi}_{\mathrm{cyl}}(\bar{w}) = +\bar{\varphi}_{\mathrm{pl}}(e^{\bar{w}}) - \frac{\bar{\alpha}_{\mathrm{pl}}}{2}\bar{w}. \tag{A.3}$$

Thus a common leading cylinder exponential comes from opposite source-plane momenta,

$$\beta_{\mathrm{pl}} = -\varepsilon\beta_1 + O(\varepsilon^3), \qquad \bar{\beta}_{\mathrm{pl}} = +\varepsilon\beta_1 + O(\varepsilon^3). \tag{A.4}$$

The barred equal-time Fourier modes are reindexed. Under inversion, the contour orientation and the Jacobian combine so that the transported one-form has no residual sign,

$$\mathrm{d}\bar{z}\,\bar{V}(\bar{z}) \longmapsto \mathrm{d}z\,\widetilde{\bar{V}}(z), \qquad \widetilde{\bar{V}}(z) = z^{-2}\bar{V}(z^{-1}). \tag{A.5}$$

This fixes the sum and difference channels in equation (4.8).

Fix $z_0 \in \mathbb{C}^{\times}$, write $z = z_0 e^{\mathrm{i}\theta}$ and $\ell = \log(z/z_0) = \mathrm{i}\theta$. Define

$$\operatorname*{LRes}_{z_0} f := \frac{1}{2\pi\mathrm{i}} \int_0^{2\pi} f(z_0 e^{\mathrm{i}\theta}) \mathrm{i} z_0 e^{\mathrm{i}\theta}\,\mathrm{d}\theta. \tag{A.6}$$

For $r \in \mathbb{Z}$ and formal a,

$$\operatorname*{LRes}_{z_0} z^{r-1} e^{a\ell} = z_0^{r} \frac{e^{2\pi\mathrm{i}a} - 1}{2\pi\mathrm{i}(r+a)}, \tag{A.7}$$

with the removable value at $r + a = 0$. In particular, $\operatorname{LRes}_{z_0} \mathrm{d}\ell = 1$, so logarithmic total derivatives may be discarded only after the exact horizontal monodromy has been imposed.

The finite A_1 family in a common fibre

The root-normalized construction of section 4 can be checked completely in the single-field normalization. Introduce the single-field branch functions $b_{\mathrm{u}}(t) = -2\mu_{\mathrm{u}}(t)$ and $b_{\mathrm{b}}(t) = -2\mu_{\mathrm{b}}(t)$. The two regular branches obey

$$\begin{aligned} t b_{\mathrm{u}}(t)^2 - 2(\alpha^+ + t\alpha^-) b_{\mathrm{u}}(t) - 4 = 0, \\ t b_{\mathrm{b}}(t)^2 - 2(\alpha^+ - t\alpha^-) b_{\mathrm{b}}(t) - 4 = 0, \end{aligned} \tag{A.8}$$

with

$$b_{\mathrm{u}}(t) = \beta_1 + t\beta_3 + O(t^2), \qquad b_{\mathrm{b}}(t) = \beta_1 + t\bar{\beta}_3 + O(t^2). \tag{A.9}$$

The transported affine one-forms are represented by

$$V_{\mathrm{u}}(z,t) = z^{tb_{\mathrm{u}}(t)^2/4} :\exp\{b_{\mathrm{u}}(t)(\varphi^+ + t\varphi^-)\}:_{\mathrm{n}}, \tag{A.10}$$

$$V_{\mathrm{b}}(z,t) = z^{tb_{\mathrm{b}}(t)^2/4} :\exp\{b_{\mathrm{b}}(t)(\varphi^+ - t\varphi^-)\}:_{\mathrm{an}} . \tag{A.11}$$

Their first Taylor coefficients, before converting the second orientation, are

$$\begin{aligned} V_{\mathrm{u}} &= E + t :\left(\beta_1\varphi^- + \beta_3\varphi^+ + \frac{\beta_1^2}{4}\ell\right) E: + O(t^2), \\ V_{\mathrm{b}} &= E + t :\left(-\beta_1\varphi^- + \bar{\beta}_3\varphi^+ + \frac{\beta_1^2}{4}\ell\right) E: + O(t^2), \qquad \ell = \log(z/z_0). \end{aligned} \tag{A.12}$$

Thus the strict half-difference has density

$$:\left(\beta_1\varphi^- + B_{\mathrm{diff}}\varphi^+\right) E:, \qquad B_{\mathrm{diff}} = \frac{\beta_3 - \bar{\beta}_3}{2} = -\frac{\alpha^-\beta_1}{\alpha^+}, \tag{A.13}$$

whereas the common parameter/affine tangent is

$$:\left(D\varphi^+ + \frac{\beta_1^2}{4}\ell\right) E:, \qquad D = \frac{\beta_3 + \bar{\beta}_3}{2} = \frac{\beta_1^2}{2\alpha^+}. \tag{A.14}$$

Using $\varphi^{\pm}_{\mathrm{sf}} = -\alpha_i \cdot \varphi^{\pm}/2$ and $\beta_1 = -2\mu$, equations (A.13) and (A.14) become exactly equations (4.10) and (4.11).

The affine powers in equations (A.10) and (A.11) also fix the moving horizontal section. For an integer monodromy label m, simultaneous periodicity requires

$$\lambda^+(t) = \frac{1}{2}\left[\frac{m - tb_{\mathrm{u}}(t)^2/4}{b_{\mathrm{u}}(t)} + \frac{m - tb_{\mathrm{b}}(t)^2/4}{b_{\mathrm{b}}(t)}\right], \tag{A.15}$$

$$t\lambda^-(t) = \frac{1}{2}\left[\frac{m - tb_{\mathrm{u}}(t)^2/4}{b_{\mathrm{u}}(t)} - \frac{m - tb_{\mathrm{b}}(t)^2/4}{b_{\mathrm{b}}(t)}\right]. \tag{A.16}$$

Substitution verifies both branch monodromy equations identically; the right-hand side of equation (A.16) vanishes at $t = 0$, so its apparent pole is removable. In the common equal-time polarization the anti-normal vertex contributes the scalar factor

$$\bar{Z}_{t,\rho} = (1-\rho)^{-tb_{\mathrm{b}}(t)^2/2}, \qquad Z_{t,\rho} = 1, \tag{A.17}$$

which is the single-field form of equation (4.4). Since it is independent of z, a finite subtraction changes the first coefficient only by a scalar multiple of the leading charge.

Finally, the target momenta of the two exact intertwiners are

$$\begin{aligned} p^+_{\mathrm{T,u}} &= \lambda^+(t) + \frac{tb_{\mathrm{u}}(t)}{4}, & p^+_{\mathrm{T,b}} &= \lambda^+(t) - \frac{tb_{\mathrm{b}}(t)}{4}, \\ p^-_{\mathrm{T,u}} &= \lambda^-(t) + \frac{b_{\mathrm{u}}(t)}{4}, & p^-_{\mathrm{T,b}} &= \lambda^-(t) + \frac{b_{\mathrm{b}}(t)}{4}. \end{aligned} \tag{A.18}$$

Taking their relative first tangent gives

$$\Delta p^+ = \frac{\beta_1}{2}, \qquad \Delta p^- = \frac{B_{\mathrm{diff}}}{2}, \tag{A.19}$$

which reproduces equation (4.14) and hence the inhomogeneous terms equations (4.17) and (4.18). This calculation is the coefficient-level bridge between the affine parent vertices, their horizontal residues and the fixed-target deformation equation.

B Explicit nonlinear connection fields

Before listing the connection fields, it is useful to record a formula-level check of the minus primary. On differential polynomials in P_a and α^+ define the directional Miura derivation

$$\mathfrak{D} = \sum_{a=1}^{2} \sum_{r\geq 0} N_a^{(r)} \frac{\partial}{\partial P_a^{(r)}} + \alpha^- \frac{\partial}{\partial \alpha^+}, \qquad P_a^{(r)} := \partial^r P_a, \quad N_a^{(r)} := \partial^r N_a. \tag{B.1}$$

After applying $\mathfrak{D}$, every monomial containing one N is put in the averaged order equation (2.8). Direct differentiation gives

$$U_2^- = \mathfrak{D} U_2^+, \qquad \mathcal{U}_3^- = \mathfrak{D} \mathcal{U}_3^+. \tag{B.2}$$

The highest-derivative term is particularly diagnostic:

$$\mathfrak{D}\left[-\frac{(\alpha^+)^2}{2} P_2''\right] = -\alpha^+\alpha^- P_2'' - \frac{(\alpha^+)^2}{2} N_2''. \tag{B.3}$$

This identity fixes the relative factor in equation (2.16); it also implies the one-N filtration used in the Wick expansion. Equivalently, it follows by differentiating the ordered parent row determinant under $(P, \alpha^+) \mapsto (P + sN, \alpha^+ + s\alpha^-)$ at $s = 0$ before the two orientations are averaged.

The shift derivatives in equation (5.2) give

$$\mathcal{A}_1 = \frac{1}{2z^3}\left[2z^2(P_1^2 - P_2^2) + 2(\alpha^+)^2 + \alpha^+(3zP_1 - z^2P_1' + zP_2 + z^2P_2')\right], \tag{B.4}$$

$$\mathcal{A}_2 = -\frac{1}{2z^3}\left[2z^2(P_1^2 + 2P_1P_2) + 2(\alpha^+)^2 + \alpha^+(3zP_1 - z^2P_1' + 2zP_2 - 2z^2P_2')\right], \tag{B.5}$$

$$\begin{aligned}\mathcal{B}_1 = \frac{1}{2z^3}\Big[&4z^2\{\mathrm{Av}(N_1P_1) - \mathrm{Av}(N_2P_2)\} + \alpha^-(3zP_1 - z^2P_1' + zP_2 + z^2P_2') \\ &+\alpha^+(3zN_1 - z^2N_1' + zN_2 + z^2N_2' + 4\alpha^-)\Big],\end{aligned} \tag{B.6}$$

$$\begin{aligned}\mathcal{B}_2 = -\frac{1}{2z^3}\Big[&4z^2\{\mathrm{Av}(N_1P_1) + \mathrm{Av}(N_1P_2) + \mathrm{Av}(N_2P_1)\} \\ &+ \alpha^-(3zP_1 - z^2P_1' + 2zP_2 - 2z^2P_2') \\ &+\alpha^+(3zN_1 - z^2N_1' + 2zN_2 - 2z^2N_2' + 4\alpha^-)\Big].\end{aligned} \tag{B.7}$$

They satisfy $\mathfrak{d}_i^- \mathcal{U}_3^- = \mathcal{A}_i$ and generate the target differences equation (5.4).

C Complete strict-odd pole coefficients

If the N-containing averaged products are replaced prematurely by commutative monomials, the corrected residuals are

$$R_1^{\mathrm{comm}} = -\frac{\mu^2(P_1 + 3P_2)}{4w} - \frac{\mu}{4w^2}, \qquad R_2^{\mathrm{comm}} = \frac{\mu^2(P_1 - 2P_2)}{4w} + \frac{\mu}{4w^2}. \tag{C.1}$$

The w^{-2} terms follow from $\partial_w(w^{-1}) = -w^{-2}$ and are not killed by horizontality. The two finite-parent orientations generate precisely equation (5.13), which cancels equation (C.1) and converts the following pole data into exact derivatives.

Writing

$$\mathrm{Sing}[\mathcal{U}_3^-(z)\mathcal{S}^{[1]}_{i,\mathrm{odd}}(w)]_{\mathrm{Av}} = \sum_{q=1}^{3} \mathcal{O}^{\mathrm{Av}}_{i,q}(w) E_i(w)(z-w)^{-q}, \tag{C.2}$$

the first-root coefficients are

$$\mathcal{O}^{\mathrm{Av}}_{1,3} = \frac{H_1 + \alpha^-}{2}, \tag{C.3}$$

$$\mathcal{O}^{\mathrm{Av}}_{1,2} = \frac{\mu H_1(3P_1 + P_2) + 3N_1 + N_2}{4}, \tag{C.4}$$

$$\mathcal{O}^{\mathrm{Av}}_{1,1} = \frac{1}{2}\Big[\mu^2 H_1(P_1^2 - P_2^2) + \mu H_1(P_1' + P_2') + 2\mu\,\mathrm{Av}(N_1P_1 - N_2P_2) + N_1' + N_2' - \alpha^-\mu^2(P_1^2 - P_2^2)\Big]. \tag{C.5}$$

For the second root,

$$\mathcal{O}^{\mathrm{Av}}_{2,3} = -\frac{H_2 + \alpha^-}{2}, \tag{C.6}$$

$$\mathcal{O}^{\mathrm{Av}}_{2,2} = -\frac{\mu H_2(3P_1 + 2P_2) + 3N_1 + 2N_2}{4}, \tag{C.7}$$

$$\mathcal{O}^{\mathrm{Av}}_{2,1} = -\frac{1}{2}\Big[\mu^2 H_2(P_1^2 + 2P_1P_2) + \mu H_2 P_1' + 2\mu\,\mathrm{Av}(N_1P_1 + N_1P_2 + N_2P_1) + N_1' - \alpha^-\mu^2(P_1^2 + 2P_1P_2)\Big]. \tag{C.8}$$

They obey

$$\mathcal{O}^{\mathrm{Av}}_{i,3} = 2\Xi_{i,2}, \qquad \mathcal{O}^{\mathrm{Av}}_{i,2} = \partial\Xi_{i,2} + \Xi_{i,1}, \qquad \mathcal{O}^{\mathrm{Av}}_{i,1} = \partial\Xi_{i,1}, \tag{C.9}$$

where the derivatives include their action on E_i. These identities reconstruct equation (5.10) pole by pole.

D Bulk coefficient solve and explicit Miura fields

At $s = 2$, the bulk invariants are

$$M_{\mathrm{bulk}} = -\frac{1}{4}(\psi')^2 - \frac{1}{2}\psi'', \qquad L_{\mathrm{bulk}} = -\frac{1}{2}\psi'\bar{\psi}' - \frac{1}{2}\bar{\psi}''. \tag{D.1}$$

Substituting the general ansatz preceding equation (6.5) into the two boundary fields, together with the crossed bracket and $D^2 = -1$, produces the independent coefficient conditions

$$2A^2 + \frac{\rho_M}{4} = 0, \qquad 2Da_2A + \frac{\rho_M}{2} = 0, \qquad 4AB + \frac{\rho_L}{2} = 0, \qquad 2Da_2B + \frac{\rho_L}{2} = 0,$$
$$4AC + \frac{\rho_{LM}}{4} = 0, \qquad 2Da_2C + \frac{\rho_{LM}}{2} = 0, \qquad D^2 + 1 = 0, \qquad AB + \frac{a_2^2}{4} = 0. \tag{D.2}$$

The solve is elementary and leaves no continuous normalization freedom. The first pair in equation (D.2) gives $\rho_M = -8A^2$ and $A = Da_2/2$; the second pair similarly gives $\rho_L = -8AB$ and $B = Da_2/2$. The crossed bracket then yields

$$AB + \frac{a_2^2}{4} = 0 \quad \Longrightarrow \quad D^2 = -1. \tag{D.3}$$

The two equations involving C imply $C = 0$ and hence $\rho_{LM} = 0$. Therefore

$$A = B = \frac{Da_2}{2}, \qquad \rho_M = \rho_L = 2a_2^2, \qquad C = \rho_{LM} = 0, \qquad D = \pm\mathrm{i}. \tag{D.4}$$

The two solutions are globally conjugate; equation (6.5) is the branch with $\alpha^+ = +\mathrm{i}a_2$.

For $s = 3$, the bulk fields appearing in equation (6.16) are obtained from

$$D_0 = \mathrm{diag}\left(\psi'_{(L)} + \frac{2}{3}\psi'_{(W)}, -\frac{4}{3}\psi'_{(W)}, -\psi'_{(L)} + \frac{2}{3}\psi'_{(W)}\right), \tag{D.5}$$

and the same expression with barred fields for D_1. The general diagonal parametrization fixes the common scalar-to-weight map

$$\boldsymbol{\psi} = -\mathrm{i}a_3 \begin{pmatrix} 1/\sqrt{2} & \sqrt{2} \\ \sqrt{6}/2 & -\sqrt{6}/3 \end{pmatrix} \begin{pmatrix} \psi_{(L)} \\ \psi_{(W)} \end{pmatrix}, \tag{D.6}$$

with the same matrix for $\bar{\boldsymbol{\psi}}$. It implies $\chi_i = (\mathrm{i}/a_3)\alpha_i \cdot \boldsymbol{\psi}$ and leads to the fields

$$M_{(L)} = \frac{1}{12}\left[-3(\psi'_{(L)})^2 - 4(\psi'_{(W)})^2 - 6\psi''_{(L)}\right], \tag{D.7}$$

$$L_{(L)} = \frac{1}{6}\left[-3\psi'_{(L)}\bar{\psi}'_{(L)} - 4\psi'_{(W)}\bar{\psi}'_{(W)} - 3\bar{\psi}''_{(L)}\right], \tag{D.8}$$

$$M_{(W)} = \frac{1}{108}\left[\psi'_{(W)}\{18(\psi'_{(L)})^2 - 8(\psi'_{(W)})^2 + 9\psi''_{(L)}\} + 27\psi'_{(L)}\psi''_{(W)} + 9\psi'''_{(W)}\right], \tag{D.9}$$

$$\begin{aligned} L_{(W)} = \frac{1}{36}\Big[&\bar{\psi}'_{(W)}\{6(\psi'_{(L)})^2 - 8(\psi'_{(W)})^2 + 3\psi''_{(L)}\} + 3\psi'_{(W)}\bar{\psi}''_{(L)} \\ &+9\psi'_{(L)}\bar{\psi}''_{(W)} + 3\bar{\psi}'_{(L)}\{4\psi'_{(L)}\psi'_{(W)} + 3\psi''_{(W)}\} + 3\bar{\psi}'''_{(W)}\Big]. \end{aligned} \tag{D.10}$$

Direct substitution of equation (6.15) into the boundary currents gives equation (6.16) coefficientwise.

E Normalization and transcription conventions

The bulk comparison uses the displayed equations and explicit $s = 2, 3$ sectors of the source papers. Three localized printed formulas require clarification.

(i) The sentence following eq. (2.16) of [1] gives $\alpha^+ = \mathrm{i}/\sqrt{2G}$. The equations immediately preceding it state $c_M = -12(\alpha^+)^2 = 3/G$, which instead require $\alpha^+ = \mathrm{i}/(2\sqrt{G})$, as do equations (6.1) and (6.5).

(ii) Equation (5.28c) of [2] prints $9(\psi''_{(L)})^2$ in $M_{(W)}$. After multiplication by $\psi'_{(W)}$ this has differential weight five. The general row determinant and the coefficient identity equation (6.16) require the linear term $9\psi''_{(L)}$ used in equation (D.9).

(iii) Equation (6.23c) of [2] displays $-[2k_{\mathrm{CS}}\tau_s]^{-1}(\delta_{ij} - 1/s)\delta'$. Reducing the paper's diagonal map to each explicit bracket at $s = 2$ and $s = 3$ gives twice that value, namely $-[k_{\mathrm{CS}}\tau_s]^{-1}(\delta_{ij} - 1/s)\delta'$. The theorem above uses the explicit $s = 2, 3$ brackets and records this disagreement rather than introducing separate field rescalings.

For the contracted free fields we use the mode bracket containing the factor m, as in eq. (3.14) of [1]; it is also the form fixed by their contracted brackets and $N = 2$ normalization. The comparisons above use these internally consistent normalizations; no other formula from the cited constructions is modified.

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